Hitesh V S

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Education

Vellore Institute of Technology — Chennai, India
B.Tech, Electrical and Electronics Engineering ***(CGPA: 8.56/10, Year 1)*** — *July 2025 – July 2029*

Leadership & Research Experience

Technical Lead — 2026 – Present
IEEE Solid-State Circuits Society (SSCS) Student Chapter, VIT Chennai — *Chennai, India*

- Lead the chapter's technical activities, organizing hands-on workshops on circuits, PCB design and embedded systems
- Mentor junior members on project ideas, simulations and hardware prototyping
- Coordinate with faculty advisors and industry speakers to host technical talks and events

Undergraduate Researcher — 2025 – Present
VIT Chennai — *Chennai, India*

- Designed an 8-bit hybrid approximate adder (ETA-1) for edge filtering, cutting switching activity 40% at 28.22 dB PSNR
- Authored a paper accepted at ICAIET and a journal manuscript under revision at *The Journal of Supercomputing*
- Developed an autonomous aquatic waste skimmer; patent cleared internally by VIT, filing in process

Publications & Patent

ICAIET Conference Paper | *ML-Driven Approximate Computing for* 3×3 *Image Blurring* — Accepted, 2027

The Journal of Supercomputing | *Error Tolerant Adder Type-1 Application in Facial Recognition* — Under revision, 2026

Patent Application | *Autonomous System for Cleaning Static Water Bodies via Sensor-Based Navigation & Debris Collection* — In process, 2026

Projects

ALPS: Autonomous Aquatic Waste Skimmer | *ESP32, GPS, Sensors, Motors* — 2025

- Designed and built an ESP32-based autonomous surface vessel that collects floating waste from water bodies
- Integrated a NEO-6M GPS module for waypoint navigation and sensors for obstacle detection

UpliftEd: Multilingual AI Learning & Safety | *Python, YOLOv8, Web Speech API, PHP, MySQL* — 2025

- Engineered a real-time e-learning and moderation platform with live speech translation in 6 languages
- Deployed a YOLOv8 vision safety guard detecting hostile behavior in live video, triggering call termination and alarms
- Built an NPTEL-style micro-credential engine with AI-generated notes, quizzes and a CustomTkinter desktop UI

Li-ion Battery Thermal Runaway Prediction | *Python, Machine Learning* — 2026

- Built a model predicting thermal runaway in Li-ion cells from NASA PCoE data, reaching 98% accuracy
- Benchmarked three algorithms and combined them into a hybrid model to improve precision and recall

Analog Irrigation System | *LabVIEW, Sensors* — 2026

- Developed an analog soil-moisture-based irrigation controller with a LabVIEW monitoring and control interface

Technical Skills

Programming: Python, C, Embedded C, MATLAB
Hardware & Tools: ESP32, Oscilloscope, Multimeter, KiCad, LTspice, Simulink, ModelSim, LabVIEW
Core Areas: Power Electronics, Control Systems, Analog Circuits, Approximate Computing, Machine Learning
Developer Tools: Git, GitHub, VS Code

Interests

Astrophotography, Space Technology and Aviation, Carnatic Music (violin and keyboard)